CONFERENCE PROGRAM

IEEE SmartGridComm™

IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids

26–29 October 2026

College Station, Texas, USA

Texas A&M Hotel & Conference Center



IEEE ComSoc
IEEE Communications Society

IEEE

TEXAS A&M UNIVERSITY
Department of Electrical & Computer Engineering

HOWDY!

At the Conference

WiFi

Network: **TBD**

Password: **TBD**

TBD — confirm WiFi details with the hotel

Registration Desk

Pre-Function I & II

Mon · Oct 26	**8:00 AM–5:00 PM**
Tue · Oct 27	**8:00 AM–5:00 PM**
Wed · Oct 28	**8:30 AM–4:00 PM**
Thu · Oct 29	**8:30 AM–12:00 PM**

All sessions take place at the **Texas A&M Hotel & Conference Center**, 177 Joe Routt Blvd, College Station, TX 77840. All times are Central Time. The program is subject to change; the online program below is always up to date.

Scan & Go

Full Program

Interactive schedule with every session, room & paper

smartgridcomm2026.github.io/SmartGridComm-2026-Program

Proceedings

Conference papers (access for registered attendees)

smartgridcomm26.ieeevtc.org

Website

News, venue & travel information

sgc2026.ieee-smartgridcomm.org

Tap a code or link to open it.

Highlights

4 keynotes · **2 industry panels** · **3 tutorials** · **90 papers** in 5 technical symposia plus **22 workshop papers** · Welcome Reception (Mon 6:00 PM) · Gala Dinner with the Aggie Wranglers (Tue 7:00 PM) · Best Paper Awards (Thu 12:30 PM) · Optional campus & technical tours (Thu 2:00 PM)

WELCOME

From the General Co-Chairs

We are delighted to welcome you to the 2026 edition of the IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), a flagship conference of the IEEE Communications Society (ComSoc), hosted this year at the Texas A&M University main campus in College Station, Texas, USA.

SmartGridComm is a premier forum for researchers and practitioners to exchange ideas, explore enabling technologies and share experiences related to smart grids. We are excited about the content that will be presented and the ideas that will be exchanged. Whether you are attending for the first time or have been a regular contributor, you will engage with a unique community of researchers, practitioners, and industry leaders that synthesize skills in power systems, distributed control, data analytics, communications, and cybersecurity.

The SmartGridComm authors and participants have consistently been crucial players in pioneering impactful solutions at the intersection of power systems, communications, control, and cybersecurity. The conference has consistently maintained its standing as a distinguished platform, renowned for its scholarly contributions. In 2026, we are thrilled to expand that legacy with a strong program of technical papers, keynotes, panels, and workshops. In today's global context, the need for ubiquitous power systems resilience is inseparable from dependable and secure communications and control. As it addresses these needs, the SmartGridComm program is essential for safeguarding society, which depends now more than ever on data, computers, artificial intelligence and of course electric power. Solutions are presented at the intersection of these layers, and we look forward to these being shared within and beyond our conference.

We express deep appreciation to those whose efforts made this edition possible, including all of the Technical Program Chairs, the Symposia Chairs, the Workshop Chairs, the Tutorial Chairs, all of the organizing committee, and the extensive cohort of TPC members and reviewers who generously devoted their time to ensure the rigor of our review process. Thank you also to our distinguished speakers, panelists, and our sponsors. We are deeply grateful. Thank you for the exceptional support of the Communications Society, especially the help of Conference Manager, Melissa Torres, Publication Chair, James Irvine, and Financial Chair, Bruce Worthman.

We warmly welcome you to College Station. With about 75,000 students, Texas A&M is one of the few American universities designated as a land, sea, and space grant university. It promotes education and research in fields as diverse as agriculture and space exploration. It serves as a premier institution for research collaborations. It is a deep honor to host you, our exemplary international community and esteemed colleagues. We eagerly anticipate your participation and to welcoming you soon.

Katherine Davis, Ana Goulart, Aditya Ashok, Manimaran Govindarasu

General Co-Chairs, IEEE SmartGridComm 2026

WELCOME

From the Technical Program Committee Co-Chairs

On behalf of the Technical Program Committee, it is our great pleasure to welcome you to the IEEE International Conference on Smart Grid Communications (SmartGridComm) 2026.

IEEE SmartGridComm remains a premier venue for researchers, practitioners, and industry leaders to exchange ideas, present cutting-edge research, and explore innovative solutions that shape the future of smart grids. This year's conference brings together a strong and dynamic community to address the challenges and opportunities in grid modernization, resilience, cybersecurity, and sustainable energy systems. The program reflects the dedication and expertise of our global contributors and the rigorous review process conducted by our esteemed reviewers and Technical Program Committee (TPC) members

For the 2026 SmartGridComm conference, we are enthusiastic to share the forthcoming technical program. We received 210 paper submissions this year: 180 for the main program consisting of 5 technical symposia, and 30 for 2 workshops. Out of these, 90 outstanding papers will be presented in main program and 22 in the workshops. In addition, the conference will feature 3 tutorials delivered by leading experts, 4 extraordinary keynotes delivered by distinguished speakers from industry and from academia, discussing recent trends in electrical power networks and digitalization. A few common themes in the technical content of the conference this year include emerging smart grid solutions and their challenges and opportunities, particularly the resilience, defense, and implications of artificial intelligence and data centers in the smart grid context, and innovations in enabling communication systems, security and privacy, and other computing and algorithmic advances. The conference program also includes 2 industry panels that will explore the future of smart grid communications. In addition to the engaging technical program, we will be pleased to announce the awarding of travel grants and best paper awards.

We would like to express our gratitude to the co-chairs of the 5 technical symposia for their dedicated support to SmartGridComm. They efficiently managed to secure at least 3 reviews for each paper within a tight timeframe, ensuring a smooth decision-making process. We also sincerely thank the workshop organizers for putting together a set of timely workshops and organizing the review process in a professional manner. We would like to thank the members of the IEEE SmartGridComm organizing committee for their dedication and support during the entire period of the technical program preparation and development. We would also like to thank the technical program committee (TPC) members for their diligent work that upholds our high standards for the technical program papers. We also sincerely thank the keynote speakers and panelists for contributing to the SmartGridComm program. Finally, we would like to thank the authors, constituting the scientific backbone of this forum, for all the precious knowledge they will share with their peers. We are looking forward to seeing you all in College Station in October!

Chenye Wu, Eirini Eleni Tsiropoulou, György Dán, Saman Zonouz

Technical Program Committee Co-Chairs, IEEE SmartGridComm 2026

PROGRAM

Program at a Glance

Monday, October 26

Day 1 · Workshops & tutorials

8:00	Main Registration Desk Open
9:00	Workshops & Tutorials
11:00	Workshops & Tutorials
12:30	Lunch & Networking
1:30	Workshops & Tutorials
4:00	Workshops & Tutorials
6:00	**Welcome Reception**

Tuesday, October 27

Day 2 · Opening, keynotes & parallel paper sessions

8:00	Registration Desk Open
8:45	Welcome & Opening Remarks
9:00	**Keynote: Dr. Vince Poor**
10:15	Parallel Technical Sessions
12:15	Lunch
1:30	**Keynote: Dr. Tom Overbye**
2:45	Parallel Technical Sessions
6:00	**Reception**
7:00	**Gala Dinner**

Wednesday, October 28

Day 3 · Keynote, industry panels & parallel paper sessions

8:30	Registration Desk Open
8:45	Day 3 Welcome
9:00	**Keynote: Dr. Karen Butler-Purry**
10:15	**Panel 1 · OPAL-RT**
12:15	Networking Lunch
1:30	**Panel 2 · G3-Alliance**
2:45	Parallel Technical Sessions

Thursday, October 29

Day 4 · Keynote, final paper sessions, awards & closing

8:30	Registration Desk Open (morning only)
8:45	Day 4 Welcome
9:00	**Keynote: Dr. Veronica Adetola**
10:15	Parallel Technical Sessions
12:30	Best Paper Awards & Closing Session
2:00	Walking Tour & Technical Tours

Full program with rooms, sessions & papers

Technical sessions run in parallel in Century I & II, Laurel, Ross and Corps. Scan or tap for every session, chair and paper.

smartgridcomm2026.github.io/SmartGridComm-2026-Program

KEYNOTE SPEAKERS

Keynotes



TUESDAY, OCTOBER 27 · 09:00–10:00 · CENTURY I & II

H. Vincent Poor

Princeton University

"Some Approaches to Power Grid Resilience"

H. Vincent Poor is the Michael Henry Strater University Professor at Princeton University, where his interests include information theory, stochastic analysis and machine learning, and their applications in wireless networks, energy systems and related fields. During 2006-2016, he served as the Dean of Princeton's School of Engineering and Applied Science, and he has also held visiting positions at a several other universities, including most recently at Berkeley, Caltech and Texas A&M. Among his publications is the book Advanced Data Analytics for Power Systems (Cambridge University Press, 2021). Dr. Poor is a member of the U.S. National Academy of Engineering and the U.S. National Academy of Sciences, and he a foreign member of the Royal Society and other national and international academies. Other recognition of his work includes the 2009 IEEE ComSoc Edwin Howard Armstrong Achievement Award and the 2017 IEEE Alexander Graham Bell Medal.



TUESDAY, OCTOBER 27 · 13:30–14:30 · CENTURY I & II

Thomas J. Overbye

Texas A&M University

"Enhancing the Resilience of Large-Scale Electric Grids"

Thomas J. Overbye is a Professor and holder of the O'Donnell Foundation Chair III in the Department of Electrical and Computer Engineering at Texas A&M University. He serves as Director of the Texas A&M Engineering Experiment Station Smart Grid Center and is a member of the National Academy of Engineering. His research and teaching focus on electric power systems, power-grid modeling and simulation, renewable-energy integration, grid resilience, and smart-grid technologies. He also serves as faculty advisor for Christian Engineering Leaders, a Texas A&M student organization.

Keynotes (continued)



WEDNESDAY, OCTOBER 28 · 09:00–10:00 · CENTURY I & II

Karen L. Butler-Purry

Texas A&M University

Karen L. Butler-Purry is a Professor in the Department of Electrical and Computer Engineering at Texas A&M University. From 2010 to 2022, she served as Associate Provost for Graduate and Professional Studies and Dean of the Graduate and Professional School. Her research focuses on electric power systems, including microgrids and power distribution in electric vehicles. She is an IEEE Fellow, elected in 2018 for contributions to expanding minority participation in power systems education. Her other honors include the AAAS Mentor Award and the Council of Graduate Schools' Debra Stewart Award for Outstanding Leadership in Graduate Education.



THURSDAY, OCTOBER 29 · 09:00–10:00 · CENTURY I & II

Veronica Adetola

Pacific Northwest National Laboratory (PNNL)

"Advancing Stability and Resilience in Power and Energy Systems: AI-Enabled Control and Co-Design"

Veronica Adetola is a Chief Research Scientist and Team Leader at Pacific Northwest National Laboratory (PNNL), where she leads multidisciplinary research to advance the efficiency, stability, and resilience of power and energy systems. Her research spans model-based and learning-enabled control, optimization, embedded and edge AI control, cyber-physical systems resilience, and system–control co-design for diverse applications, including buildings, microgrids, and bulk power systems. She has held technical leadership roles in PNNL initiatives advancing resilient control and multi-objective energy-system co-design for emerging grid challenges, including the reliable integration of large electric loads. Before joining PNNL in 2019, Adetola spent nearly a decade conducting industry research at United Technologies Research Center, now Raytheon Technologies Research Center. She has authored a book, published more than 60 peer-reviewed journal and conference papers, and is an inventor on 11 issued U.S. patents. She holds a Ph.D. in engineering with a specialization in control systems.

Panels

PANEL 1 · ORGANIZED BY OPAL-RT · WEDNESDAY, OCTOBER 28 · 10:15 AM – 12:15 PM · CENTURY I & II

Understanding the Role of Hardware-in-the-Loop Testing in Supporting Reliable Data Center Interconnections and Grid Operations

The rapid growth of AI-powered applications has fueled unprecedented demand to build and power new data centers. System operators are seeing a tremendous volume of new data center interconnection applications and are adapting both their requirements and their study processes to understand how data centers will affect an already evolving bulk power grid. This panel brings together system operators, data center developers and owners, researchers, and vendors to discuss model validation, the key role of hardware-in-the-loop testing in reliable interconnection and operation, and the regulatory and operational drivers behind it.

Aditya Ashok & Zerui Dong — OPAL-RT Corporation

Ashish Upreti — Google — Data center owner's perspectives on model validation and the use of HIL testing tools

Zerui Dong — OPAL-RT Corporation — Derisking data center design, testing, and grid integration through high-fidelity modeling and real-time simulations

Billy Yancey — Electric Power Engineers — Practical challenges and regulatory gaps in model validation of data centers

PANEL 2 · ORGANIZED BY G3-ALLIANCE · WEDNESDAY, OCTOBER 28 · 1:30 – 2:30 PM · CENTURY I & II

Innovation in Smart Grid Connectivity and AI towards AMI 2.0

A joint presentation followed by an open discussion with the audience.

Riccardo Fiorelli — STMicroelectronics

Kaveh Razazian — Sagemcom

Social Events

Welcome Reception · Monday, Oct 26 · 6:00–8:00 PM

Block T Bar & Grill, Texas A&M Hotel & Conference Center.

Reception & Gala Dinner · Tuesday, Oct 27 · 6:00–10:00 PM

Century I & II Ballroom. An evening of dinner, conversation and Texas tradition featuring the **Aggie Wranglers**, Texas A&M's country & western dance team, followed by a SmartGridComm 2027 preview. Included in registration.

Optional Tours · Thursday, Oct 29 · 2:00–3:30 PM

Campus walking tour, followed by a technical tour of the Global Cyber Research Institute (GCRI) in the Blocker Building.

VENUE

Texas A&M Hotel & Conference Center

177 Joe Routt Boulevard, College Station, TX 77840 · on the Texas A&M University campus.

FLOOR PLAN GOES HERE

Request the meeting-space floor plan from the hotel (sales@texasamhotelcc.com) or ComSoc, save it in assets/, and put its filename in floor_plan.

Rooms

Century I & II	Keynotes, panels, opening & closing, gala dinner, Tue–Thu lunches; technical sessions (Data Analytics & Computation); Cyber-Physical Resilience workshop
Laurel	Technical sessions (Control & Operation); Tutorial 2
Ross	Technical sessions (Communications & Networking, joint sessions, Control & Operation); Tutorial 3
Corps	Technical sessions (Cyber-Physical Security & Privacy); Tutorial 1; Digital Twin workshop
Pre-Function I & II	Registration desk, Monday lunch, Tuesday reception
Block T Bar & Grill	Monday welcome reception (in the hotel)

LOCAL GUIDE

Explore Aggieland

Make the most of your time in Bryan–College Station with some of the area's most popular attractions.

George H.W. Bush Presidential Library and Museum

On the Texas A&M campus, the museum explores the life and presidency of the 41st President through historical exhibits, documents, artifacts and interactive displays.

Texas A&M University Campus

See some of the university's best-known landmarks, including Kyle Field, the Century Tree and the Bonfire Memorial. The university offers a self-guided campus tour for visitors.

The Gardens at Texas A&M University

A relaxing walk through Texas A&M's public gardens and green spaces on West Campus.

Century Square

A destination near campus with restaurants, shops, outdoor spaces and entertainment.

Historic Downtown Bryan

Locally owned restaurants and shops, art galleries, historic architecture and live entertainment in an official Texas Cultural District.

Getting Around

Easterwood Airport (CLL)

About 8 minutes from the hotel, with daily nonstop American Airlines service to Dallas/Fort Worth (DFW).

Rideshare & Taxi

Uber, Lyft and taxis operate throughout Bryan–College Station. The Ground Shuttle connects Houston (IAH); Hitch offers city-to-city rides from Austin.

On Campus

Texas A&M's AggieSpirit buses run around campus; routes and live tracking are available through Texas A&M Transportation Services and apps such as Maroon Rides.

COMMITTEES

Organizing Committee

GENERAL CHAIR

Katherine Davis, Associate Professor, Electrical & Computer Engineering, Texas A&M University

GENERAL CO-CHAIR

Aditya Ashok, Director of Energy Systems at OPAL-RT Corporation

Manimaran Govindarasu, Professor, Electrical and Computer Engineering, Iowa State University

Ana E. Goulart, Associate Professor, Engineering Technology, Texas A&M University

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Chenye Wu, School of Science and Engineering, The Chinese University of Hong Kong

Eirini Eleni Tsiropoulou, School of Electrical, Computer and Energy Engineering, Arizona State University

Saman Zonouz, Electrical & Computer Engineering, Georgia Institute of Technology

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Chenye Wu, School of Science and Engineering, The Chinese University of Hong Kong

Eirini Eleni Tsiropoulou, School of Electrical, Computer and Energy Engineering, Arizona State University

György Dán, School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology

Saman Zonouz, Electrical & Computer Engineering, Georgia Institute of Technology

SYMPOSIUM: DATA ANALYTICS AND COMPUTATION

Mia Naeini, Electrical Engineering, University of South Florida

Xin Chen, Electrical & Computer Engineering, Texas A&M University

Hongjian Sun, Department of Engineering, Durham University

SYMPOSIUM: CONTROL AND OPERATION

Olugbenga Moses Anubi, Electrical & Computer Engineering, Florida State University

Ali Mehrizi-Sani, Electrical & Computer Engineering, Virginia Tech

Li Zhengmao, Aalto University

SYMPOSIUM: COMMUNICATIONS AND NETWORKING

Anurag Srivasta, Computer Science and Electrical Engineering, West Virginia University

Fabrizio Granelli, Information Engineering and Computer Science, University of Trento

Atul Bohara, Relativity

SYMPOSIUM: CYBER-PHYSICAL SECURITY AND PRIVACY

Daisuke Mashima, Information Systems Technology and Design, Singapore University of Technology and Design

Qinghua Li, Computer Science & Computer Engineering, University of Arkansas

Subhash Laksminarayana, School of Engineering, University of Warwick

SYMPOSIUM: POWER LINE COMMUNICATIONS AND ITS APPLICATIONS

Lutz Lampe, Electrical and Computer Engineering, University of British Columbia

Riccardo Fiorelli, Smart Grid Application Manager, STMicroelectronics

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G3-Alliance

Enabling the smartest grid...together

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